Department of Mathematics and Statistics

Assignment #3, MAS2001

Course Cover: Theory of Estimation: Maximum Likelihood and method moment estimation, Sufficient statistics, Bayesian estimation, Confidence intervals for means.

SECTION A	
A1	The criteria for good estimators are ________.
A2	Write any two methods of estimation _______.
A3	If M.L.E. exists, it is the most _______ in the class of such estimators.
A4	Estimate the parameter p of the binomial distribution by the method of moments.
A5	Let T_1 and T_2 be statistics with expectations $E(T_1) = q_1 + q_2$ and $E(T_2) = q_1 - q_2$. Find unbiased estimators of q_1 and q_2.
A6	Under what circumstances can the normal distribution be used to find confidence limits of the populations mean?
A7	The interval ____ is known as the 95% confidence interval for m , and the 95% confidence limits are
A8	Under Bayesian approach, the parameter of distribution is a _______.
A9	Bayes estimators are always a function of ________ statistics.
A10	Expain (i) Unbiasedness (ii) Consistency (iii) Efficiency (iv) Sufficiency
SECTION B	
B1	Let $x_1, x_2, \dots, x_n$ be a random sample from $N(\mu, \sigma^2)$ population. Find sufficient estimators for μ, and σ^2.
B2	Find the maximum likelihood estimate for the parameter λ of a Poisson distribution. On the basis of sample of size n.
B3	In random sampling from normal population $N(\mu, \sigma^2)$, find the maximum likelihood estimators for (i) μ when σ^2 is known (ii) σ^2 when μ is known

B4	A quality control expert is required to estimate the mean thickness of aluminum sheets used in the production of airframes. A sample of 100 sheets revels a mean of 0.048 inches with a standard deviation of 0.01 inches. Construct the 99 per cent confidence interval. Ans. (0.0454, 0.0506)
B5	Show that in estimating for the parameter in the Poisson distribution with general term $\frac{e^{-\mu}}{x!}\mu^x$, $\bar{x}$ is sufficient for μ.

SECTION-C	
C1	Let $x_1 = 0.7, x_2 = 1.5, x_3 = 0.3, x_4 = 0.5, x_5 = 3.4$ be a sample of size 5 from a distribution having pdf $f(x, \theta) = \theta e^{-\theta x}, x > 0, \theta > 0$. Find the MLE of θ.
C2	Let, $x_1, x_2, \ldots, x_n$ be a random sample of n observations from a population having pdf $f(x, \theta) = \theta e$ ^{$-\theta$x}, $x > 0$. The prior distribution of θ is $f_1(\theta) = 1/\theta, 0 < \theta < \infty$. Find the Bayes estimator of θ.
C3	A random sample of size 10 was drawn from a normal population with an unknown mean and a variance of 44.1 inch^2. If the observations are (in inches): 65,71,80,76,78,82, 68,72,65 and 81 obtain the 95% confidence interval for the population mean. Ans. (69.68, 77.92) inches
C4	For a $N(\mu, \sigma^2)$ population with variance equal to 9, Construct a 95% confidence interval for the parametric function $r(\theta) = 2\mu + 7$, based on the random sample size 25, giving a sample mean of 30. (tabulated value=1.96).
C5	Let X be a binomial random variable with parameters n and p. where $f_1(p) = 1, 0 < p < 1$. Find the Bayes estimator of p.

	SECTION-D
D1	X_1, X_2, and X_3 is a random sample of size 3 from a population with mean value μ and variance σ^2. T_1, T_2, T_3 are the estimators used to estimate mean value μ, where $$T_1 = X_1 + X_2 - X_3, T_2 = 2X_1 + 3X_3 - 4X_2 \text{ and } T_3 = \frac{1}{3}(\lambda X_1 + X_2 + X_3)$$ (i) Are T_1 and T_2 unbiased estimators? (ii) Find the value of λ such that T_3 is unbiased estimators for μ.

	(iii) With this value of λ is T_3 a consistent estimator? (iv) Which is the best estimator?
D2	A random sample of 50 sales invoices was taken from a large population of sales invoices. The average value was found to be Rs 2000 with a standard deviation of Rs 540. Find a 90 per cent confidence interval for the true mean value of all the sales. (tabulated value=1.645).
D3	The personnel department of an organization would like to estimate the family dental expenses of its employees to determine the feasibility of providing a dental insurance plan. A random sample of 10 employees reveals the following family dental expenses (in thousand Rs) in the previous year: 11, 37, 25, 62, 51, 21, 18, 43, 32, 20. Set up a 99 per cent confidence interval of the average family dental expenses for the employees of this organization. (tabulated value=1.833).
D4	survey conducted by a shopping mall group showed that a family in a metro-city spends an average of Rs 500 on cloths every month. Suppose a sample of 81 families resulted in a sample mean of Rs 540 per month and a sample standard deviation of Rs 150, develop a 95 per cent confidence interval estimator of the mean amount spent per month by a family (tabulated value=1.96)